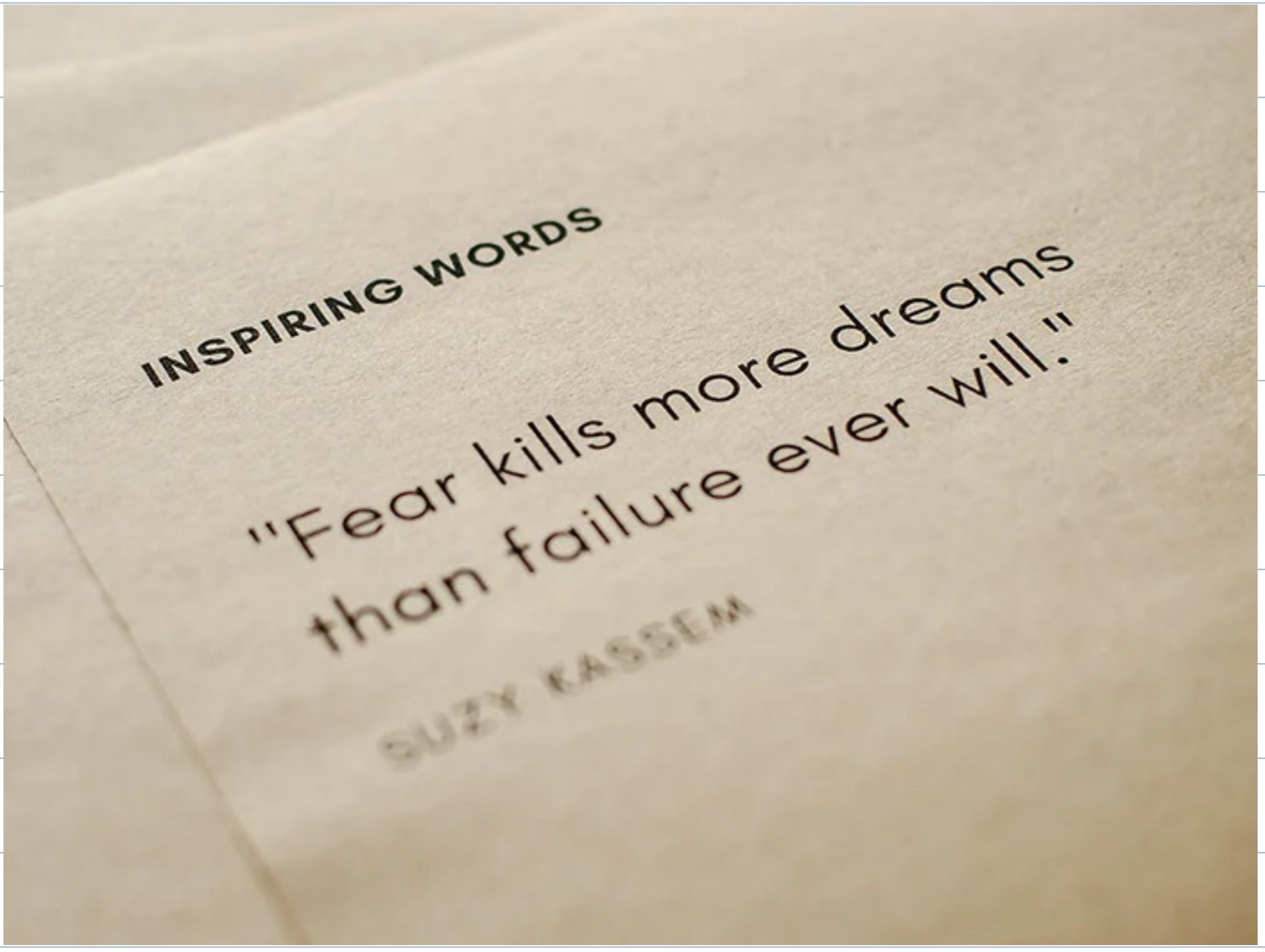


Agenda:

1. Search in row wise col wise sorted matrix
2. Sum of all sub-matrices sum
3. First Missing Integer
4. First Missing Integer for ve- nos.

Search in row-wise & column-wise sorted 2D array

$$\begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix}$$

N * M

Search (6) → true

Search (15) → false

Bruteforce → ∀ i, j check if A[i][j] == K.

TC = O(N * M)   SC = O(1)

Observation

increasing →   ← decreasing

$$\begin{array}{c|c} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix} \end{array}$$

increasing ↓ (left, rows 0–1) · decreasing ↑ (left, rows 2–3)

increasing ↓ (right, rows 0–1) · decreasing ↑ (right, rows 2–3)

increasing →   ← decreasing

decreasing $\leftarrow$

Search (6)

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & & & & \\ 1 & & & & \\ 2 & & & & \\ 3 & & & & \end{array} \quad \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix}$$

$\downarrow$ increasing

$6 < 13 \quad 6 > 1 \quad 6 > 3$

$6 > 5 \quad 6 < 10$

decreasing $\leftarrow$

Search (4)

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & & & & \\ 1 & & & & \\ 2 & & & & \\ 3 & & & & \end{array} \quad \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix}$$

$\downarrow$ increasing

Ans = false

×

\# time we go left = $M-1$

\# times we go down = $N-1$

\# checks = $(N-1) + (M-1) + 1 = N+M-1$

-5	-2	1	13
-4	0	3	14
-3	2	6	18

```
r = 0      c = M-1
while (r < N && c >= 0) {
    if (A[r][c] == K) return true
    if (A[r][c] > K)  c--
    else   r++
}
return false
```

TC = $\underline{O(N+M)}$

SC = $\underline{O(1)}$

<u>Submatrix</u> →

continuous part of matrix.

$$
\begin{array}{c c}
 & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\
\begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix}
\end{array}
$$

Q → Given a matrix, find sum of all submatrix sum.

$$A = \begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix}$$

$[4] \to 4$ $\quad [4\ 9] \to 13$ $\quad \begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix} \to 25$

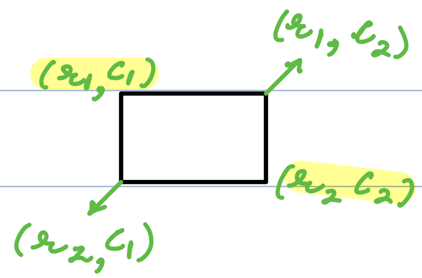
$[9] \to 9$ $\quad [9\ 6] \to 15$ $\quad \begin{bmatrix} 4 \\ 5 \end{bmatrix} \to 9$ $\quad \begin{bmatrix} 9 & 6 \\ -1 & 2 \end{bmatrix} \to 16$

$[6] \to 6$ $\quad [5\ -1] \to 4$

$[5] \to 5$ $\quad [-1\ 2] \to 1$ $\quad \begin{bmatrix} 9 \\ -1 \end{bmatrix} \to 8$ $\quad \begin{bmatrix} 4 & 9 \\ 5 & -1 \end{bmatrix} \to 17$

Ans = 166 $\quad [-1] \to -1$ $\quad [4\ 9\ 6] \to 19$

$[2] \to 2$ $\quad [5\ -1\ 2] \to 6$ $\quad \begin{bmatrix} 6 \\ 2 \end{bmatrix} \to 8$



Bruteforce

```
ans = 0
for i → 0 to (N-1) {
    for j → 0 to (M-1) {
        for k → i to (N-1) {
            for l → j to (M-1) {
                sum = 0
                for r → i to k {
                    for c → j to l {
                        sum += A[r][c]
                    }
```

}

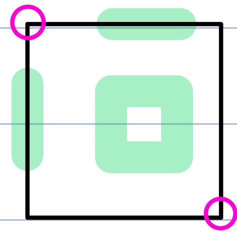
ans += sum

} } } }

return ans

$TC = \underline{O(N^3 * M^3)}$ $\quad SC = \underline{O(1)}$

# of all submatrix →



\# subarrays of row → $\frac{N*(N+1)}{2}$

\# subarrays of col → $\frac{M*(M+1)}{2}$

$\frac{N*(N+1)}{2} * \frac{M*(M+1)}{2}$

Contribution Technique → $Ans = \sum_{\forall i,j}$ contribution of $A[i][j]$

$A[i][j]$ * (# submatrix it is present)

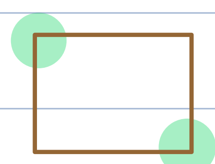
Find # submatrix containing $A[2][1]$.

$$
\begin{array}{cc}
 & \begin{array}{cccc} 0 & 1 \; (j) & 2 & 3 \end{array} \\
\begin{array}{c} 0 \\ 1 \\ i \;\; 2 \\ 3 \end{array} &
\begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & \boxed{2} & 5 & 18 \\ 2 & 6 & 10 & 20 \end{bmatrix}
\end{array}
$$

\# top left → 6

\# bottom right → 6

$6*6 = \underline{36}$



$[0 \quad i] \rightarrow i+1 \qquad [0 \quad j] \rightarrow j+1$

top left cells $= (i+1) * (j+1)$

bottom right cells $= (N-i) * (M-j)$

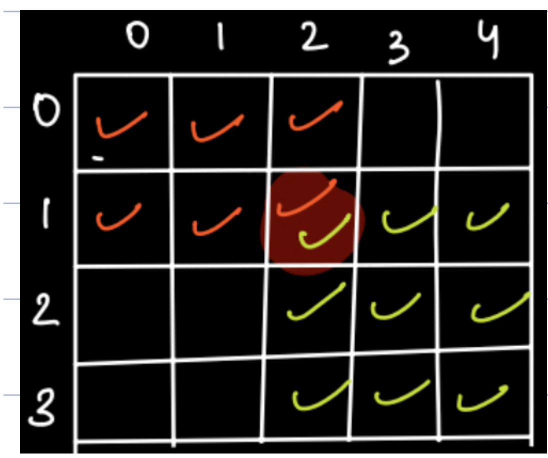
$[i \quad N-1] \rightarrow N-1-i+1 = N-i$

$[j \quad M-1] \rightarrow M-1-j+1 = M-j$

submatrix $= ((i+1) * (j+1)) * ((N-i) * (M-j))$

```
ans = 0
for i → 0 to (N-1) {
    for j → 0 to (M-1) {
        ans += A[i][j] * (i+1)*(j+1) * (N-i) * (M-j)
    }
}  return ans
```

TC = O(N*M)   SC = O(1)



$6 * 9 = \underline{54}$

Q → Find first missing +ve integer.

$$A = [\overset{0}{3} \quad \overset{1}{8} \quad \overset{2}{6} \quad \overset{3}{1}] \qquad Ans = \underline{2}$$

$$A = [3 \quad -9 \quad 2 \quad 0 \quad -1 \quad 6] \qquad Ans = \underline{1}$$

$$A = [5 \quad 3 \quad 1 \quad -1 \quad -2 \quad -4 \quad 7 \quad 2] \qquad Ans = \underline{4}$$

$$A = [1 \quad 2 \quad 3 \quad 4 \quad 5] \qquad Ans = 6$$

Bruteforce → min ans = 1

max ans = N + 1

$\forall i \rightarrow [1 \quad N+1]$, check if it is present in A[].

$TC = \underline{O(N^2)}$ $\qquad SC = \underline{O(1)}$

[_ _ _ _ _] Ans → [1 6]

Sol 1 → If $\forall i$, A[i] is positive

$$\begin{array}{cccccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ A = [& \cancel{3} & \cancel{5} & \cancel{6} & 1 & \cancel{2} & \cancel{9} & 10 & 5 & \cancel{1} \;] \\ & -3 & -1 & -6 & & -2 & -9 & & & -1 \end{array}$$

Ans = $\underline{4}$

check if [1 N] is present.

if A[i] → [1 N] → mark A[i] as present

index → [0 N-1]   ↳ update index (A[i] − 1) → *(−1)

```
for i → 0 to (N-1) {
    x = |A[i]|
    if ( 1 <= x  &&  x <= N)
        A[x-1] = (-1) * |A[x-1]|
    }
}

for i → 0 to (N-1) {
    if (A[i] >= 0)
        return i+1
}

return (N+1)
```

TC = $\underline{O(N)}$ SC = $\underline{O(1)}$

<u>Sol 2</u> → Mark vst → update element to its correct position in sorted order.

	1	2	3	4	5	6	7	8	9
	✓	✓	✓	✓	✓	✓	✓	✓	✓
	0	1	2	3	4	5	6	7	8
A =	[~~3~~	~~0~~	~~6~~	~~1~~	~~2~~	~~−4~~	10	~~5~~	−2]
	~~6~~	2	3	−4	~~0~~	6		0	
	~~−4~~ 1	✓	✓		5				

Ans = 4

HW → Try code for the above.